Sadguru Balumama Shikshan Prasarak Mandal's
K.P.PATIL INSTITUTE OF TECHNOLOGY
(DTE CODE : 6814) (MSBTE Code : 1661)
Approved by AICTE, DTE Mumbai &Govt. of Maharashtra,Affiliated to MSBTE Mumbai

Internships Daily Diary(2025-26)
Department Of Artificial Intelligence & Machine Learning Engineering

Program Code :AI5K
Name of the Student : Sanika Dhanaji chougale
Name of the Mentor (Faculty) : Mrs. P.M. Kumbhar
Name of the Mentor (Industry): Mr. Abhijit Gatade
Enrollment Number : 23213500044

Week	Day & Date	Discussion Topics/Activity	Details of Work Allotted Till Next Session /Corrections Suggested/Faculty Remarks	Signature of Industry Mentor
Week 8	Mon, Date 21/07/25	♦ Project introduction and data set study	Studied heart disease dataset & features like cp, thalach, etc	
	Tue, Date 22/07/25	♦ Data cleaning & Preprocessing	Handled nulls, endoded categorical data, Scaled numericals	
	Wed, Date 23/07/25	♦ EDA & Visualization	Created plots checked correction, reviewed class balance	IGAP TECHNOLOGIES PVT.LTD. Kolhapur
	Thu, Date 24/07/25	♦ Model training	Trained LR, DT, KNN, RF using K-fold and evaluated metrics	
	Fri, Date 25/07/25	♦ Model tuning and Saving	Tuned models with GridSearch cv, saved final RF model.	
	Sat, Date 26/07/25	♦ APP planing & future Scope	Designed Streamlit app, listed inputs, planned improvements	

Internships Daily Diary (2025-26)

Week No: 08 **Day:** Monday **Date:** 21/07/25

"Project Introduction & Dataset Study"

On the first day, I was introduced to the project topic focused on heart disease prediction.

I began by exploring the heart disease dataset, understanding its structure and identifying the key features used for prediction. Special attention was given to critical variables such as chest pain type (cp), maximum heart rate (thalach), and others.

This foundational step helped set the direction for data preprocessing and model development in the upcoming days.

Internships Daily Diary (2025-26)

Week No: 08 **Day:** Tuesday **Date:** 22/07/25

"Data Cleaning & Preprocessing"

I worked on cleaning the dataset to ensure it was ready for analysis and modeling. Tasks included handling missing values, encoding categorical variables into numerical format using techniques like label encoding or one-hot encoding, and applying scaling methods (such as Standard Scaler or MinMax Scaler) to bring all numerical features into a uniform range. This step was essential for improving model performance and reliability.

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Internships Daily Diary (2025-26)

Week No: 08 **Day:** Wednesday **Date:** 23/07/25

"EDA and Visualization"

EDA was conducted to understand patterns, relationships, and anomalies within the dataset. Using libraries such as Matplotlib and Seaborn, I created various various plots like histograms, bar charts, and correlation heatmaps.

These visualizations helped in identifying trends, checking feature distributions, and examining class imbalance, which was crucial for choosing appropriate models and evaluation strategies. This helped me choose the rights models later.

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Internships Daily Diary (2025-26)

Week No: 08 **Day:** Thursday **Date:** 24/07/25

"Model Training"

On this day, I implemented and trained several classification models, including Logistic Regression (LR), Decision Tree (DT), K-Nearest Neighbors (KNN), and Random forest (RF).

I used K-fold cross-Validation to evaluate model stability and performance, Comparing metrics such as accuracy, percision recall, and f1-score. This provided a strong foundation for Selecting the best-performing model.

K-fold used in any others accuracy, and other metrics.

Internships Daily Diary (2025-26)

Week No: 08 **Day:** friday **Date:** 25/07/25

"Model Tuning and Saving"

To optimize model performance, I performed hyperparameter tuning using GridSearch CV. After testing different parameter Combinations, I selected the best-performing Random forest model. The final model was saved for deployment or further use. This process enhanced the accuracy and generalizability of the project outcomes.

I also documented the best parameters and model details for future reference.

Overall, the day focused on improving, finalizing, and storing the best machine learning model.

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Internships Daily Diary (2025-26)

Week No: 08 **Day:** Saturday **Date:** 26/07/25

"Application Planning & future Scope"

I planned a simple application that would use the final trained model to predict heart disease risk. I listed the required input features and designed a basic layout of how the user interface would work. I also wrote down ideas for future improvements such as adding real-time input, deploying the model using a web app, and improving the app's usability for healthcare professionals or patients.

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